

SL Math School: Random Growth Models, etc.

Tutorial 3: Cameron Martin Theorem I

1 Tutorial 3: Problems

1.1 Core Problems

Problem 1 (Finite-dimensional Gaussian change of variables). Let $X = X_1, \dots, X_N$ be independent Gaussian random variables with

$$X_i \sim \mathcal{N}(0, \sigma_i^2),$$

and let $a_1, \dots, a_N \in \mathbb{R}$ be deterministic. Define

$$Y_i = X_i + a_i.$$

Show that the law of $Y = (Y_1, \dots, Y_N)$ is absolutely continuous with respect to the law of $(X_1, \dots, X_N)$, with Radon–Nikodym derivative

$$\frac{d\mathbb{P}_Y}{d\mathbb{P}_X}(x_1, \dots, x_N) = \exp \left(\sum_{i=1}^N \frac{a_i x_i}{\sigma_i^2} - \frac{1}{2} \sum_{i=1}^N \frac{a_i^2}{\sigma_i^2} \right).$$

Problem 2 (Cameron–Martin formula for piecewise linear shifts). Let B be a standard Brownian motion on $[0, 1]$. Let

$$0 = t_0 < t_1 < \dots < t_m = 1$$

be a partition of $[0, 1]$. Let $h : [0, 1] \rightarrow \mathbb{R}$ be deterministic, continuous, satisfy $h(0) = 0$, and be linear on each interval $[t_i, t_{i+1}]$. Write

$$v_i = \frac{h(t_{i+1}) - h(t_i)}{t_{i+1} - t_i}, \quad i = 0, \dots, m-1.$$

Thus v_i is the slope of h on $[t_i, t_{i+1}]$.

1. Let

$$\Delta_i B = B(t_{i+1}) - B(t_i).$$

Show that $\Delta_0 B, \dots, \Delta_{m-1} B$ are independent Gaussian random variables with

$$\Delta_i B \sim \mathcal{N}(0, t_{i+1} - t_i).$$

2. Show that

$$\Delta_i(B + h) = \Delta_i B + v_i(t_{i+1} - t_i).$$

3. Using the finite-dimensional Gaussian change of variables formula, show that for every bounded measurable function $F : \mathbb{R}^{m+1} \rightarrow \mathbb{R}$,

$$\mathbb{E}[F((B + h)(t_0), \dots, (B + h)(t_m))] = \mathbb{E}[F(B(t_0), \dots, B(t_m)) D_h(B)],$$

where

$$D_h(B) = \exp \left(\sum_{i=0}^{m-1} v_i (B(t_{i+1}) - B(t_i)) - \frac{1}{2} \sum_{i=0}^{m-1} v_i^2 (t_{i+1} - t_i) \right).$$

In other words, shifting Brownian motion by the piecewise linear function h changes the law by the exponential factor

$$D_h(B) = \exp \left(\int_0^1 h'(t) dB(t) - \frac{1}{2} \int_0^1 |h'(t)|^2 dt \right),$$

where the two integrals are understood here as finite sums, since h' is piecewise constant.

Optional extension. Explain why the same formula should hold for functionals depending on the values of B at arbitrary times in $[0, 1]$.

Hint. Condition on the values $B(t_0), \dots, B(t_m)$. Given these values, Brownian motion is made of independent Brownian bridges on the intervals $[t_i, t_{i+1}]$. Since h is linear on each such interval, adding h shifts the endpoints but does not change the bridge fluctuations inside the interval.

In words, shifting Brownian motion by the piecewise linear function h costs the exponential factor $D_h(B)$. This is the Cameron–Martin formula in the special case where h is piecewise linear.

Optional extension. Explain why the same formula should remain true for functionals depending on the values of B at arbitrary times $s_1, \dots, s_k \in [0, 1]$.

Remark: The above is a specific case of the more general Cameron-Martin Theorem. Indeed for any $\mathcal{C}^1$ function h with $h(0) = 0$, we have

$$\frac{d\mathbb{P}_{B+h}}{d\mathbb{P}_B}(B) = \exp \left(\int h'(t) dB_t - \frac{1}{2} \int |h'|^2 \right).$$

If you do not know what dB_t means, do not worry about it.

1.2 Extra Problems

Problem 3 (Small ball estimate around a piecewise linear path). Let $f : [0, 1] \rightarrow \mathbb{R}$ be a piecewise linear function satisfying $f(0) = 0$. Suppose

$$0 = t_0 < t_1 < \dots < t_n = 1$$

is a partition of $[0, 1]$ such that

$$f'(x) = a_j \quad \text{for } x \in [t_{j-1}, t_j], \quad j = 1, \dots, n.$$

Let B be a standard Brownian motion with $B(0) = 0$. Recall in the last tutorial, we showed for all $\epsilon > 0$ small enough,

$$\mathbb{P} \left(\sup_{x \in [0, 1]} |B(x)| \leq \epsilon \right) \geq c \exp \left(\frac{d \log \epsilon}{\epsilon^2} \right).$$

Show that, for all $\epsilon > 0$ small enough,

$$\begin{aligned} \mathbb{P} \left(\sup_{x \in [0, 1]} |B(x) - f(x)| \leq \epsilon \right) &\geq c \exp \left(-\epsilon \left(|a_n| + \sum_{j=1}^{n-1} |a_{j+1} - a_j| \right) \right) \\ &\quad \times \exp \left(-\frac{1}{2} \int_0^1 |f'(x)|^2 dx \right) \exp \left(\frac{d \log \epsilon}{\epsilon^2} \right). \end{aligned}$$

Remark: You may use Problem 2 from the previous tutorial.